HANIKA ANGLICAN INTEGRATED POLYTECHNIC (HAIP) · Data Mining and Data Warehousing · FA1 Group Work

OLAP Operations Demo

Drill-down, roll-up, slice, dice and pivot - with SQL and results

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Project: University Library Analytics Data Warehouse

Group: **Group X** (replace with your group number)

Members: Member 1 (Data Architect) · Member 2 (ETL Specialist) · Member 3 (Analytics & Security Lead)

Database: SQLite 3.53.1 · ETL: Python 3 + pandas · Dashboards: Power BI Desktop

Submission: November 2, 2026 · Presentation: November 6, 2026

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1. The OLAP cube

OLAP (On-Line Analytical Processing) means looking at the same data from different angles and at different levels of detail. Our "cube" has three main dimensions:

Cube dimension	Table	Hierarchy / members
Time	`dim_date`	Year → Quarter → Month → Day, and Day → Week → Month (plus academic year, semester)
Department	`dim_department`	Faculty → Department
Service	the three fact tables	Book Loans, Digital Downloads, Room Bookings

and further dimensions per service (resource type, user type, room, time slot, book category, student level). The view **`v_usage_cube`** (script 04) stacks the three fact tables on the shared dimensions so that one query can compare all services:

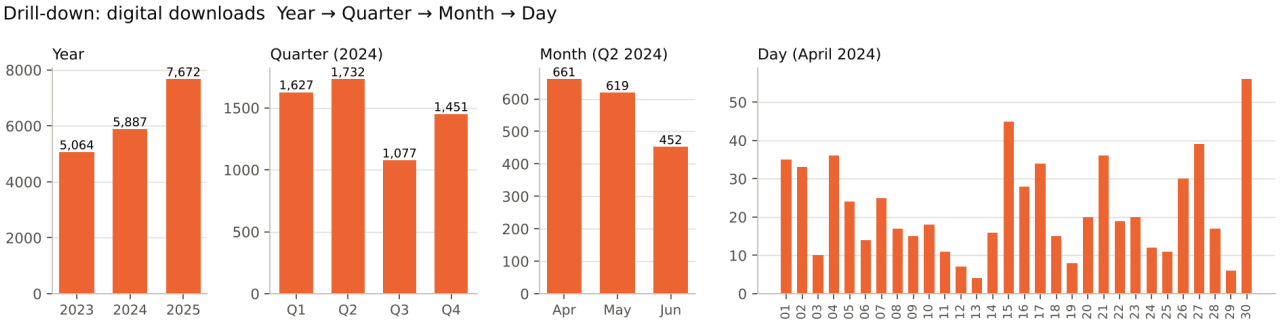
```
CREATE VIEW v_usage_cube AS
SELECT checkout_date_key AS date_key, department_key, student_key,
       'Book Loans' AS service, loan_count AS usage_count FROM fact_book_loan
UNION ALL
SELECT date_key, department_key, NULL, 'Digital Downloads', download_count FROM
fact_digital_usage
UNION ALL
SELECT date_key, department_key, student_key, 'Room Bookings', booking_count FROM
fact_room_booking;
```

How to reproduce every result in this document: open `03_Database_Files/library_dw.db` in *DB Browser for SQLite* → tab *Execute SQL* → paste the query → Ctrl+Return; or run `python run_business_queries.py` (all results go to `query_results.xlsx`). In Power BI, the same operations are done by clicking (see `dashboard_guide.pdf`, section 5).

Operation	In one sentence	Query
Drill-down	from summary to detail (Year → Day)	Q06a-d
Roll-up	from detail to summary (Day → Month)	Q07a-d
Slice	fix ONE dimension to one value	Q08a-b
Dice	filter SEVERAL dimensions at once	Q09a-b
Pivot	turn rows into columns (matrix)	Q10a-b

2. Drill-down: Year → Quarter → Month → Day

Business question 6. We start with the yearly totals and go one level deeper each time, always keeping the same measure. We follow 2024 → Q2 (exam season) → April.



Level 1 - Year (Q06a)

```
SELECT d.year_num,
       SUM(CASE WHEN u.service = 'Book Loans'        THEN u.usage_count ELSE 0 END) AS
book_loans,
       SUM(CASE WHEN u.service = 'Digital Downloads' THEN u.usage_count ELSE 0 END) AS
digital_downloads,
       SUM(CASE WHEN u.service = 'Room Bookings'     THEN u.usage_count ELSE 0 END) AS
room_bookings
FROM v_usage_cube u
JOIN dim_date d ON d.date_key = u.date_key
GROUP BY d.year_num
ORDER BY d.year_num
```

year_num	book_loans	digital_downloads	room_bookings
2023	858	5064	847
2024	721	5887	948
2025	626	7672	1042

Level 2 - Quarters of 2024 (Q06b) - the only change: `WHERE d.year_num = 2024` and `GROUP BY d.year_num, d.quarter_name`.

year_num	quarter_name	book_loans	digital_downloads	room_bookings
2024	Q1	220	1627	246
2024	Q2	196	1732	292
2024	Q3	115	1077	153
2024	Q4	190	1451	257

Level 3 - Months of Q2 2024 (Q06c) - `WHERE ... AND d.quarter_num = 2`, `GROUP BY d.month_label`.

month_label	month_name	book_loans	digital_downloads	room_bookings
2024-04	April	78	661	125
2024-05	May	66	619	100
2024-06	June	52	452	67

Level 4 - Days of April 2024 (Q06d) - `WHERE d.month_label = '2024-04'`, `GROUP BY d.full_date`.

full_date	day_name	book_loans	digital_downloads	room_bookings
2024-04-01	Monday	3	35	5
2024-04-02	Tuesday	3	33	4
2024-04-03	Wednesday	5	10	2
2024-04-04	Thursday	5	36	9
2024-04-05	Friday	4	24	3
2024-04-06	Saturday	0	14	4

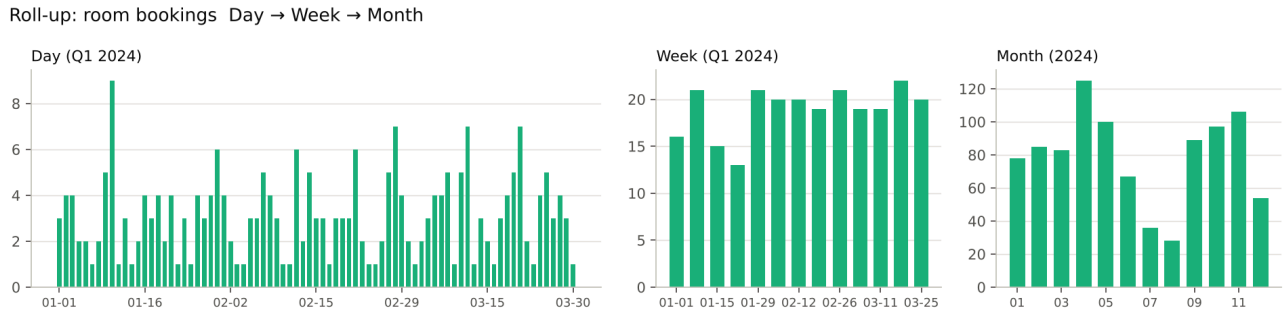
full_date	day_name	book_loans	digital_downloads	room_bookings
2024-04-07	Sunday	0	25	2
2024-04-08	Monday	4	17	5
2024-04-09	Tuesday	2	15	6
2024-04-10	Wednesday	2	18	4
2024-04-11	Thursday	2	11	9
2024-04-12	Friday	4	7	2

(first 12 of 30 rows)

Reading. Digital downloads grow every year; within 2024 the third quarter (July-August holidays) is the quietest, and April - the exam month - is the busiest month of Q2.

3. Roll-up: Day → Week → Month

Business question 7. The opposite of drill-down: daily room bookings are added up to weeks and months.



Weekly summary (Q07b)

```
SELECT d.week_start_date,
       COUNT(*)                          AS bookings,
       ROUND(SUM(f.duration_hours), 1)   AS hours_booked,
       COUNT(DISTINCT d.full_date)       AS days_with_bookings
FROM fact_room_booking f
JOIN dim_date d ON d.date_key = f.date_key
WHERE d.year_num = 2024 AND d.quarter_num = 1
GROUP BY d.week_start_date
ORDER BY d.week_start_date
```

week_start_date	bookings	hours_booked	days_with_bookings
2024-01-01	16	37.5	6
2024-01-08	21	53.5	6
2024-01-15	15	40.5	5
2024-01-22	13	32	5
2024-01-29	21	52	7
2024-02-05	20	38	7
2024-02-12	20	56.5	6
2024-02-19	19	38	7
2024-02-26	21	56.5	6
2024-03-04	19	40	6

week_start_date	bookings	hours_booked	days_with_bookings
2024-03-11	19	38.5	6
2024-03-18	22	52	6

(first 12 of 13 rows)

Monthly summary 2024 (Q07c)

month_label	bookings	hours_booked	avg_bookings_per_active_day
2024-01	78	194.5	3.1
2024-02	85	200.5	3
2024-03	83	187	3.2
2024-04	125	299.5	4.5
2024-05	100	246.5	3.8
2024-06	67	166	2.7
2024-07	36	82.5	1.7
2024-08	28	65	1.8
2024-09	89	219.5	3.9
2024-10	97	214.5	3.5
2024-11	106	239.5	4.1
2024-12	54	128	2.2

All levels in one result (Q07d). MySQL has `GROUP BY ... WITH ROLLUP`; SQLite does not, so we write the three levels and glue them with `UNION ALL`. Week rows add up to the MONTH TOTAL rows, and the months add up to the GRAND TOTAL.

```
WITH q1 AS (
    SELECT d.month_label, d.week_start_date, f.duration_hours
    FROM fact_room_booking f
    JOIN dim_date d ON d.date_key = f.date_key
    WHERE d.year_num = 2024 AND d.quarter_num = 1
)
SELECT * FROM (
    SELECT month_label, week_start_date, 'week' AS level,
           COUNT(*) AS bookings, ROUND(SUM(duration_hours), 1) AS hours_booked
    FROM q1 GROUP BY month_label, week_start_date
    UNION ALL
    SELECT month_label, NULL, 'MONTH TOTAL',
           COUNT(*), ROUND(SUM(duration_hours), 1)
    FROM q1 GROUP BY month_label
    UNION ALL
    SELECT NULL, NULL, 'GRAND TOTAL',
           COUNT(*), ROUND(SUM(duration_hours), 1)
    FROM q1
)
ORDER BY month_label IS NULL, month_label, week_start_date IS NULL, week_start_date
```

month_label	week_start_date	level	bookings	hours_booked
2024-01	2024-01-01	week	16	37.5
2024-01	2024-01-08	week	21	53.5
2024-01	2024-01-15	week	15	40.5
2024-01	2024-01-22	week	13	32
2024-01	2024-01-29	week	13	31

month_label	week_start_date	level	bookings	hours_booked
2024-01	nan	MONTH TOTAL	78	194.5
2024-02	2024-01-29	week	8	21
2024-02	2024-02-05	week	20	38
2024-02	2024-02-12	week	20	56.5
2024-02	2024-02-19	week	19	38
2024-02	2024-02-26	week	18	47
2024-02	nan	MONTH TOTAL	85	200.5

(first 12 of 19 rows)

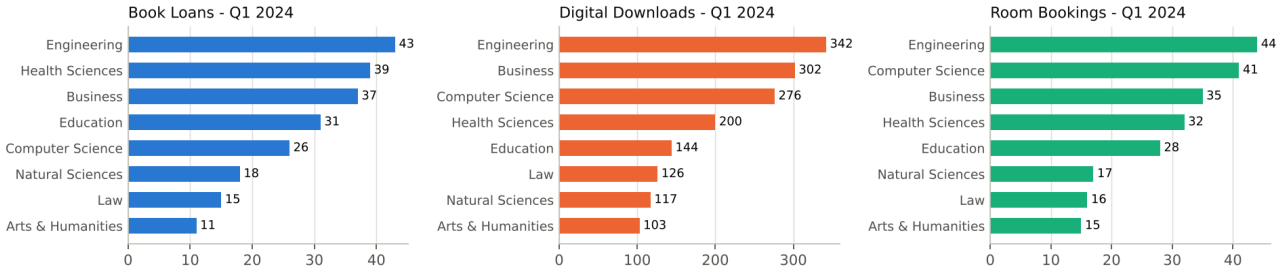
4. Slice: one time period only (Q1 2024)

Business question 8. A slice fixes **one** dimension to one value - here *time = Q1 2024* - and shows the rest of the cube.

```
SELECT u.service,
       SUM(u.usage_count)                AS total_usage,
       COUNT(DISTINCT u.date_key)        AS active_days,
       ROUND(1.0 * SUM(u.usage_count) / COUNT(DISTINCT u.date_key), 1) AS
avg_per_active_day
FROM v_usage_cube u
JOIN dim_date d ON d.date_key = u.date_key
WHERE d.year_num = 2024 AND d.quarter_name = 'Q1'          -- <== the slice
GROUP BY u.service
ORDER BY u.service
```

service	total_usage	active_days	avg_per_active_day
Book Loans	220	76	2.9
Digital Downloads	1627	89	18.3
Room Bookings	246	79	3.1

Slice: time fixed to Q1 2024



The same slice by department (Q08b)

department_name	book_loans	digital_downloads	room_bookings
Arts & Humanities	11	103	15
Business	37	302	35
Computer Science	26	276	41
Education	31	144	28
Engineering	43	342	44
Health Sciences	39	200	32

department_name	book_loans	digital_downloads	room_bookings
Law	15	126	16
Natural Sciences	18	117	17
Unknown	0	17	18

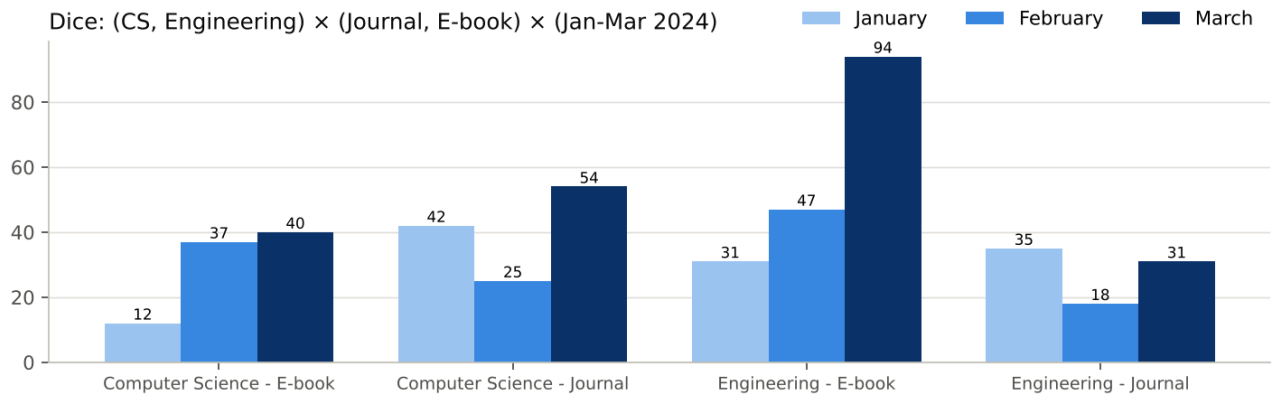
5. Dice: several dimensions at once

Business question 9 - "Engineering students' digital downloads in March" (department = ENG **and** month = March 2024 **and** user group = Student):

```
SELECT r.resource_type,
       ut.user_type,
       SUM(f.download_count)               AS downloads,
       ROUND(AVG(f.duration_minutes), 0)   AS avg_reading_minutes
FROM fact_digital_usage f
JOIN dim_date d        ON d.date_key = f.date_key
JOIN dim_department dp ON dp.department_key = f.department_key
JOIN dim_resource r    ON r.resource_key = f.resource_key
JOIN dim_user_type ut  ON ut.user_type_key = f.user_type_key
WHERE dp.department_code = 'ENG'                -- dimension 1: department
  AND d.month_label = '2024-03'                 -- dimension 2: time
  AND ut.user_group = 'Student'                 -- dimension 3: user type
GROUP BY r.resource_type, ut.user_type
ORDER BY downloads DESC
```

resource_type	user_type	downloads	avg_reading_minutes
E-book	Undergraduate Student	64	70
Article	Graduate Student	30	102
E-book	Graduate Student	16	77
Journal	Undergraduate Student	13	78
Article	Undergraduate Student	10	90
Journal	Graduate Student	7	68

A bigger dice (Q09b): (CS, Engineering) × (Journal, E-book) × (January-March 2024):



department_name	resource_type	month_name	downloads
Computer Science	E-book	January	12
Computer Science	E-book	February	37
Computer Science	E-book	March	40
Computer Science	Journal	January	42

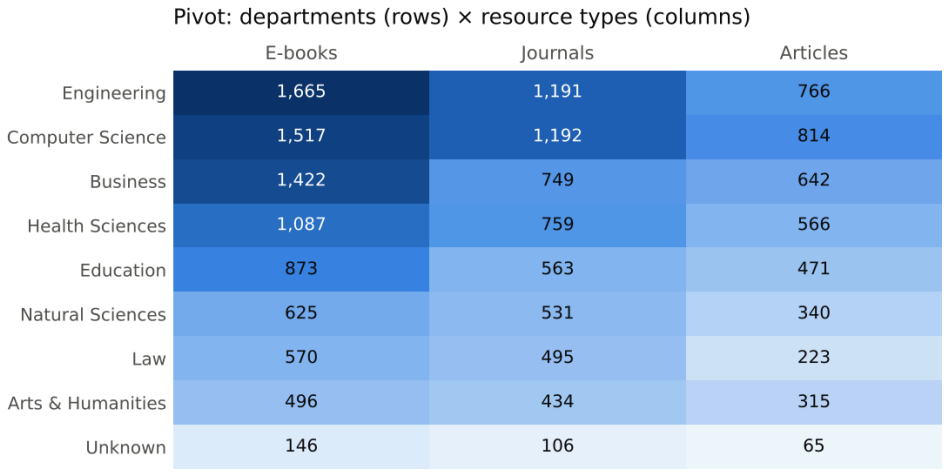
department_name	resource_type	month_name	downloads
Computer Science	Journal	February	25
Computer Science	Journal	March	54
Engineering	E-book	January	31
Engineering	E-book	February	47
Engineering	E-book	March	94
Engineering	Journal	January	35
Engineering	Journal	February	18
Engineering	Journal	March	31

6. Pivot: resource types across departments

Business question 10. A pivot turns the values of one dimension (resource type) into **columns**; departments stay as rows. In SQL this is done with `SUM(CASE WHEN ... )`; in Power BI with the *Matrix* visual.

```
SELECT dp.department_name,
       SUM(CASE WHEN r.resource_type = 'E-book'  THEN f.download_count ELSE 0 END) AS
e_books,
       SUM(CASE WHEN r.resource_type = 'Journal' THEN f.download_count ELSE 0 END) AS
journals,
       SUM(CASE WHEN r.resource_type = 'Article' THEN f.download_count ELSE 0 END) AS
articles,
       SUM(f.download_count)                                                       AS
row_total,
       ROUND(100.0 * SUM(CASE WHEN r.resource_type = 'Journal' THEN f.download_count ELSE 0
END)
             / SUM(f.download_count), 1)                                           AS
journal_share_pct
FROM fact_digital_usage f
JOIN dim_department dp ON dp.department_key = f.department_key
JOIN dim_resource r    ON r.resource_key = f.resource_key
GROUP BY dp.department_name
ORDER BY row_total DESC
```

department_name	e_books	journals	articles	row_total	journal_share_pct
Engineering	1665	1191	766	3622	32.9
Computer Science	1517	1192	814	3523	33.8
Business	1422	749	642	2813	26.6
Health Sciences	1087	759	566	2412	31.5
Education	873	563	471	1907	29.5
Natural Sciences	625	531	340	1496	35.5
Law	570	495	223	1288	38.4
Arts & Humanities	496	434	315	1245	34.9
Unknown	146	106	65	317	33.4



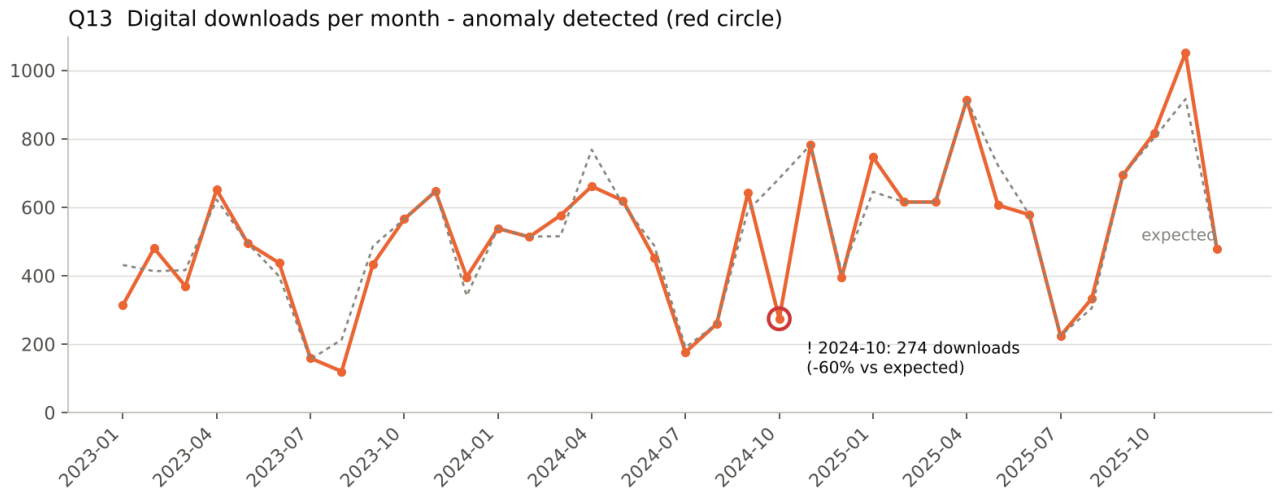
Reading. E-books are the most downloaded type in every department. Engineering and Computer Science download the most in absolute numbers, but the *share* of journals is highest in Law (38%) and lowest in Business (27%) - useful when the library decides which journal subscriptions to renew.

Second pivot (Q10b): services per year by department - shows that every department borrows fewer books in 2025 than in 2023 but downloads more:

department_name	loans_2023	loans_2025	downloads_2023	downloads_2025	rooms_2023	rooms_2025
Arts & Humanities	46	43	375	511	41	73
Business	132	91	800	1050	93	157
Computer Science	150	109	915	1387	144	158
Education	94	84	616	695	94	103
Engineering	171	104	954	1541	167	186
Health Sciences	132	104	615	1104	138	147
Law	65	45	340	553	56	79
Natural Sciences	68	46	340	688	49	63

7. Advanced analytics (bonus questions 11-13)

Question	SQL query	Python (`advanced_analytics.py`)	Result
11 Correlation	Q11: Pearson r on department-month pairs	scatter chart, r with and without the calendar effect	r = 0.341: moderate positive link
12 Forecast	Q12: linear trend for downloads	trend × seasonal pattern for all three services	2026: ≈ 8,461 downloads, 482 loans, 1,083 bookings
13 Anomaly	Q13: seasonal index vs other years	expected value = trend × median season	only **2024-10** flagged (-57.3%)



month_label	downloads	seasonal_index	normal_index	change_pct	status
2023-01	314	0.74	1.13	-34.3	normal
2023-02	480	1.14	1	13.3	normal
2023-03	369	0.87	1.07	-18.1	normal
2023-04	651	1.54	1.39	11.1	normal
2023-05	494	1.17	1.1	6	normal
2023-06	437	1.04	0.91	13.5	normal
2023-07	159	0.38	0.35	6.3	normal
2023-08	120	0.28	0.52	-45.8	normal
2023-09	433	1.03	1.2	-14.3	normal
2023-10	565	1.34	0.92	45.8	normal
2023-11	646	1.53	1.62	-5.4	normal
2023-12	396	0.94	0.78	20.7	normal

(first 12 of 36 rows)